

CYBER SECURITY

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CYBER WINTER

- WEEK 1 – INTRODUCTION TO CYBERSECURITY AND ETHICAL HACKING WITH SYLLABUS DISTRIBUTION
- WEEK 2 – NETWORKING FUNDAMENTALS FOR ETHICAL HACKING
- WEEK 3 – INTRODUCTION TO FOOTPRINTING AND RECONNAISSANCE
- WEEK 4 – SCANNING AND ENUMERATION TECHNIQUES
- WEEK 5 – SYSTEM AND NETWORK EXPLOITATION BASICS WITH WIRELESS ATTACKS
- WEEK 6 – WEB APPLICATION SECURITY FUNDAMENTALS , REMEDIATIONS AND REPORTING

WHAT IS NETWORKING?



Networking is the process of connecting computers and other devices (like phones, printers, and servers) so they can share information and resources.

Example: Imagine a group of friends in a room. When they talk, they're sharing information with each other. Networking is like that, but for computers and other devices—they're "talking" to share data and resources.

Purpose of Networking

- **Communication:** Networking enables devices to send and receive data. Just like you might send a message to a friend, one computer can send data to another.
- **Resource Sharing:** Networking lets multiple devices access shared resources, like a printer, files, or the internet. This makes it easier for people to collaborate and share information.

Relatable Example:

Think of the internet you use daily to check social media, watch videos, or read news. The internet itself is a massive network that connects millions of devices around the world, allowing us to share and access information on a global scale.

Why is Networking Important in Ethical Hacking?

Foundational Knowledge

Networking is the backbone of the internet and all online systems and it is essential to understand networking for anyone who all are looking to work in cybersecurity or ethical hacking.

How Hackers Exploit Networks

Hackers often look for weaknesses in networks to gain unauthorized access. For instance, if a network isn’t secure, a hacker might be able to intercept data or break into systems to steal information.

Ethical Hacking

In ethical hacking, our goal is to protect networks from these kinds of attacks. To defend a network effectively, we must understand how it operates and where the weak points might be.



Types of Networks We Use Daily

Home Networks

Example: When you connect your smartphone, laptop, or smart TV to your home Wi-Fi, you're creating a simple network. All your devices communicate with each other through your router and can access the internet.

Office Networks

Example: In an office, computers are often connected to a local network. This allows employees to share resources (like printers) and files with each other quickly and securely

The Internet

The internet is a huge, global network that connects millions of smaller networks and devices. It's like a web of connections that allows us to access information from anywhere in the world.



KEY NETWORKING CONCEPTS

Devices (Computers, Routers, Switches)

These are the main components that make up a network. Each has a specific role in helping devices communicate.

01 **Computer:** This is where the data (such as a file or message) starts or ends.

02 **Router:** Like a traffic cop for the internet, the router directs data where it needs to go within or outside a network.

03 **Switch:** Helps connect multiple devices within a single network (like computers in an office).



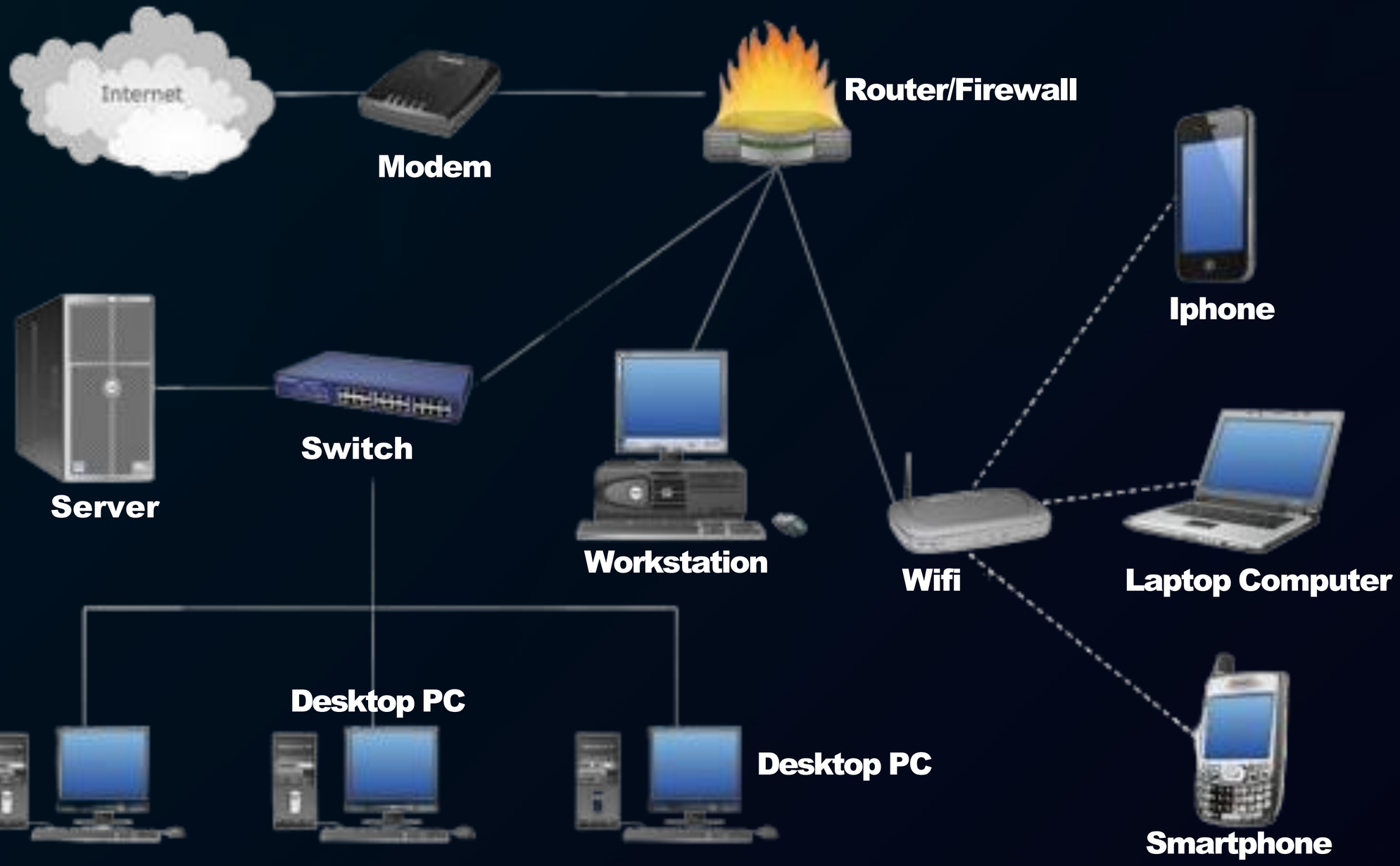
Communication

In a network, communication happens when data moves from one device to another. For example, sending an email or sharing a photo with a friend over Wi-Fi involves communication across networks.

Resources

In networking, resources refer to shared files, printers, or internet connections that everyone on a network can access.

BASIC WORKING OF NETWORK



KEY TERMINOLOGY IN NETWORKING

- **IP Address (Internet Protocol Address):** A unique identifier assigned to devices on a network. There are two versions: IPv4 (e.g., 192.168.1.1) and IPv6 (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
- **MAC Address (Media Access Control Address):** A unique hardware identifier assigned to network interface cards (NICs) of devices, allowing devices to be identified on a local network.
- **Router:** A device that routes data packets between networks, connecting different networks and directing traffic by choosing the best path for data to travel.
- **Switch:** A networking device that connects devices within the same network by forwarding data only to the device for which the data is intended, improving network efficiency.
- **Subnet:** A smaller network within a larger network, created by dividing an IP address into network and host parts. Subnetting improves network organization and security.
- **DNS (Domain Name System):** A system that translates human-readable domain names (like www.example.com) into IP addresses, allowing users to access websites without needing to remember IP addresses.
- **Firewall:** A security device or software that monitors and filters incoming and outgoing network traffic based on predefined security rules, acting as a barrier between trusted and untrusted networks.
- **Protocol:** The maximum amount of data that can be transmitted over a network connection in a given amount of time, typically measured in Mbps (megabits per second) or Gbps (gigabits per second).
- **Bandwidth:** The practice of encoding data to protect it from unauthorized access. Includes encryption and decryption.
- **Latency:** The time it takes for data to travel from the source to the destination. Lower latency means quicker communication between devices.

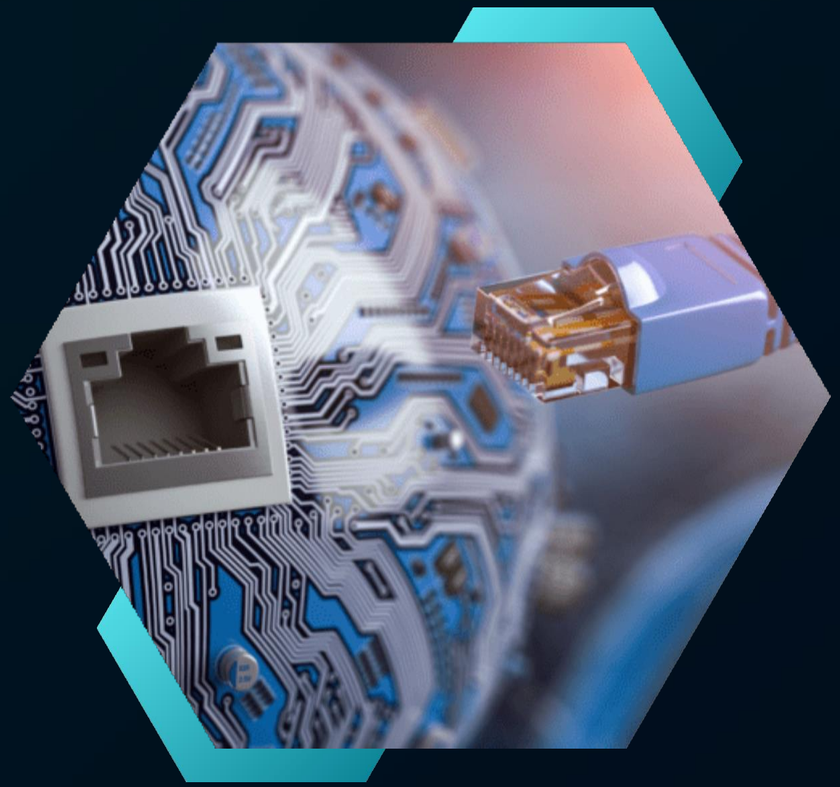
KEY TERMINOLOGY IN NETWORKING

- **Packets:** Small units of data transmitted over a network. Data is broken into packets for efficient transfer, with each packet containing source, destination, and payload information.
- **VPN (Virtual Private Network):** A service that encrypts internet traffic and routes it through a remote server, providing privacy and security when accessing public networks.
- **LAN (Local Area Network):** A network covering a small geographical area, like a home, office, or building, typically using high-speed connections like Ethernet.
- **WAN (Wide Area Network):** A network that covers a large geographical area, such as a city, country, or the internet itself, connecting multiple LANs.
- **Malware:** Malicious software designed to harm or exploit systems. Includes viruses, worms, Trojans, spyware, ransomware, and adware.
- **NAT (Network Address Translation):** A method used by routers to modify IP address information in packets, allowing multiple devices on a private network to share a single public IP address.
- **Gateway:** A device that acts as a "gate" between networks, allowing data to flow between different network types or protocols.
- **TCP/IP (Transmission Control Protocol/Internet Protocol):** The foundational communication protocol suite for the internet and most networks, responsible for end-to-end data delivery and addressing.
- **SSL/TLS (Secure Sockets Layer / Transport Layer Security):** Protocols that provide encryption and secure data transmission, primarily used to secure web communications.
- **DHCP (Dynamic Host Configuration Protocol):** A protocol that automatically assigns IP addresses to devices on a network, reducing the need for manual configuration.
- **Ping:** A utility that tests the reachability of a device on a network and measures the round-trip time of data packets.

KEY TERMINOLOGY IN NETWORKING

- **Topology:** The arrangement or structure of a network. Common types include star, bus, ring, and mesh topologies, each with unique layouts and uses.
- **OSI Model (Open Systems Interconnection Model):** A conceptual framework that standardizes the functions of a networking system into seven layers (Physical, Data Link, Network, Transport, Session, Presentation, and Application). It helps guide the design and understanding of network interactions.
- **SSID (Service Set Identifier):** The unique name of a Wi-Fi network that allows devices to identify and connect to it. It's the name users see when searching for available networks.
- **ISP (Internet Service Provider):** A company that provides individuals and organizations with internet access. Examples include Comcast, AT&T, and Verizon.
- **Port:** A logical access point for network communication. Common port numbers include 80 for HTTP, 443 for HTTPS, and 25 for SMTP (email).
- **ICMP (Internet Control Message Protocol):** A network protocol used for error messages and diagnostic functions, like the ping command. It helps troubleshoot connectivity issues.
- **Traceroute:** A diagnostic tool that traces the path of packets from source to destination, showing each network hop. Useful for identifying network bottlenecks.
- **ARP (Address Resolution Protocol):** A protocol that maps an IP address to a device's MAC address within a local network, allowing communication between devices on the same LAN.
- **Broadcast:** A network transmission sent to all devices in a particular network segment. It is often used to locate devices or deliver messages to all network members.
- **Multicast:** A method of sending data to multiple, specific devices on a network rather than to all devices, often used for streaming media to multiple users.
- **Packet Sniffing:** A symmetric encryption algorithm used for securing data. Widely adopted for its efficiency and security.

LAN / LOCAL AREA NETWORK



A Local Area Network (LAN) is a network that connects computers and devices within a small geographic area, like a single building, room, or group of rooms.

01 **Range:** Limited to a single building or a close set of rooms, like a home or office floor.

02 **Speed:** Typically fast, as it doesn't cover a large area, reducing the distance data has to travel.

03 **Security:** Easier to secure because of its small size and limited number of devices.

04 **Management:** Usually managed by one organization, such as a company or home user.

MAN / METROPOLITAN AREA NETWORK



A Metropolitan Area Network (MAN) covers a larger area than a LAN but is smaller than a WAN. It typically spans a city or a large campus, connecting multiple LANs within that area.

01 **Range:** Covers a city or large campus, connecting multiple locations within a larger area.

02 **Speed:** Generally slower than a LAN but faster than a WAN, as the distance is still relatively small.

03 **Usage:** Often used by large organizations, universities, or city governments to connect buildings and locations within a limited area.

WAN / WIDE AREA NETWORK



A Wide Area Network (WAN) spans a large geographic area, often across cities, countries, or continents. It connects multiple LANs and MANs to create a vast network like The Internet, Corporate WAN.

01 **Range:** Covers large distances—across cities, countries, and even continents

02 **Speed:** Typically slower than LANs or MANs due to the vast distances data has to travel.

03 **Complexity:** Requires complex infrastructure and is more vulnerable to security threats because it spans a large area.

04 **Usage:** Often managed by telecommunications companies who provide connections for organizations or public use.

CAN / CAMPUS AREA NETWORK



A Campus Area Network (CAN) is a type of network that covers multiple buildings within a limited area, like a university or business campus.

01 **Range:** Larger than a LAN but limited to a specific campus or organization, like a university or corporate headquarters.

02 **Speed and Efficiency:** High-speed connection is typical because it supports many users within a limited area.

03 **Usage:** Designed for users in a specific location, with resources tailored to their needs (e.g., academic databases at a university).

PAN / PERSONAL AREA NETWORK



A Personal Area Network (PAN) is the smallest type of network, covering only a few meters around a person. It connects personal devices for short-range communication like Bluetooth Devices (Wireless mouse and keyboards)

01 **Range:** Very short range, typically within a few meters.

02 **Device Connection:** Designed for personal devices that need to communicate with each other without requiring the internet or a larger network.

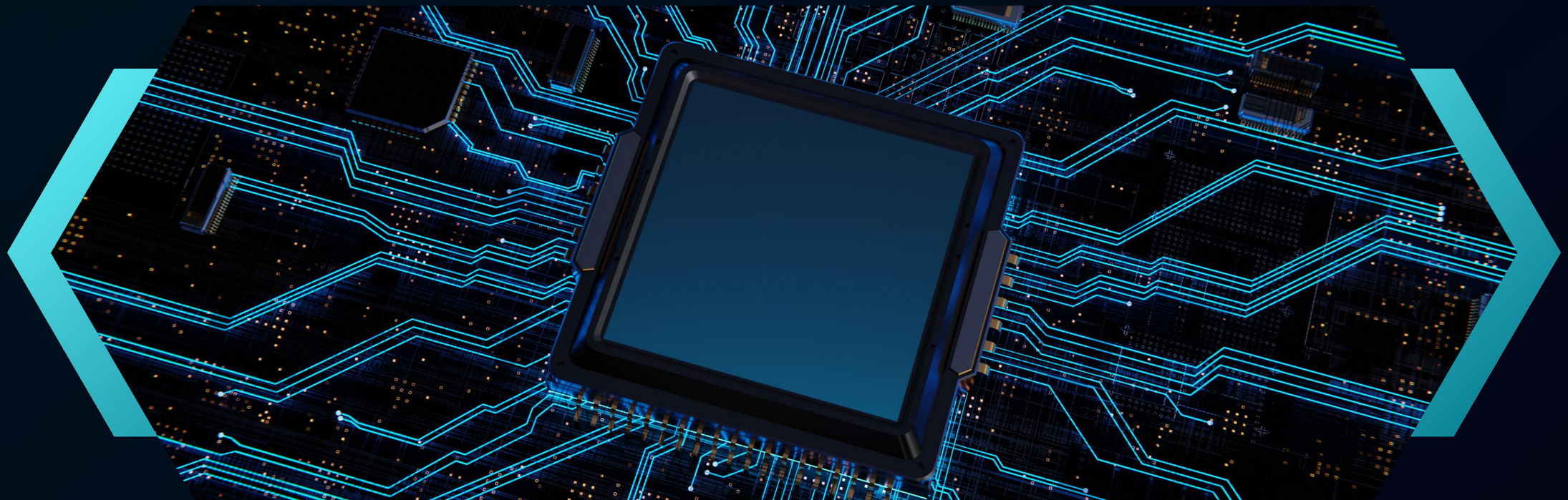
03 **Usage:** Primarily for personal use and quick, short-range connections. Often uses Bluetooth or other wireless technologies.

OSI AND TCP/IP MODEL

Why Learn About Networking Models?

01 They provide a standardized framework for understanding how data is transmitted across a network. They break down complex processes into simple layers, each with specific roles.

02 Think of these models as a recipe for baking a cake, with each step necessary for a successful outcome. Similarly, each layer in the OSI and TCP/IP models has a specific job to ensure data is transferred smoothly from one device to another.



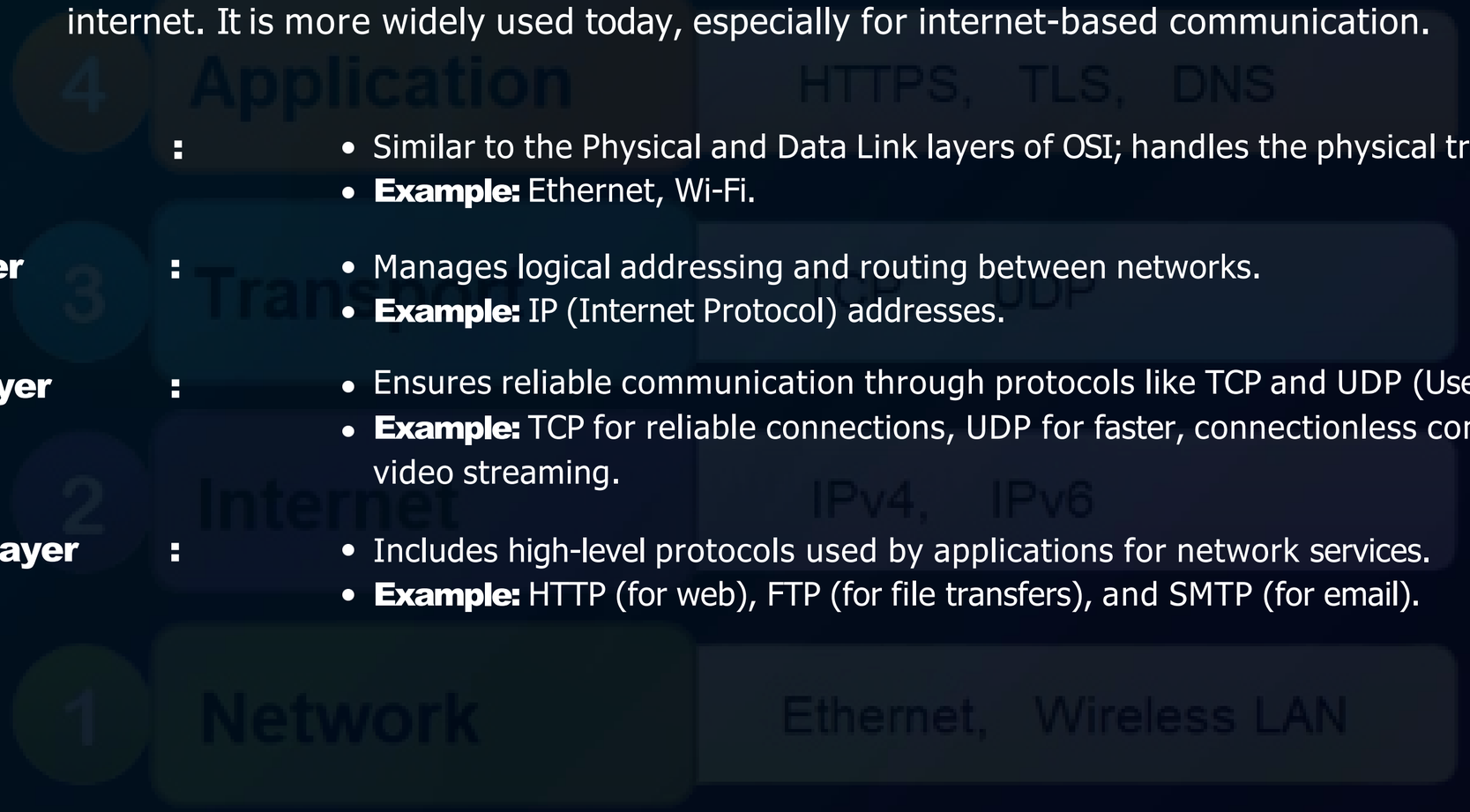
OSI MODEL OVERVIEW

The OSI (Open Systems Interconnection) model is a seven-layer framework used to understand and standardize the functions of a network.

01	**Physical Layer (LL)** :	• Manages the Physical connection between Devices. • **Example:** Cables, switches, and the physical transmission of data in the form of bits.
02	**Data Link Layer (LL)** :	• Responsible for error detection and correction, transfer of data on the same network. • **Example:** Ethernet, Wi-Fi.
03	**Network Layer (LL)** :	• Directs data between different networks and handles logical addressing, such as IP addresses. • **Example:** Routers and IP addressing.
04	**Transport Layer** :	• Ensures reliable data transfer by managing error checking and data flow control. • **Example:** TCP (Transmission Control Protocol).
05	**Session Layer (UL)** :	• Manages sessions or connections between applications. Also dialog control and synchronization. • **Example:** A secure login session for an online banking site.
06	**Presentation Layer (UL)** :	• Converts data into a format that the application layer can understand, handling encryption and decryption, Compression. • **Example:** Encryption during a secure connection.
07	**Application Layer (UL)** :	• Provides network services directly to end-users and applications. • **Example:** Web browsers, email clients.

TCP/IP MODEL OVERVIEW

The TCP/IP (Transmission Control Protocol/Internet Protocol) model is a simpler, four-layer model used as the foundation of the internet. It is more widely used today, especially for internet-based communication.



01 **Link Layer** :
- Similar to the Physical and Data Link layers of OSI; handles the physical transmission of data.
- **Example:** Ethernet, Wi-Fi.

02 **Internet Layer** :
- Manages logical addressing and routing between networks.
- **Example:** IP (Internet Protocol) addresses.

03 **Transport Layer** :
- Ensures reliable communication through protocols like TCP and UDP (User Datagram Protocol).
- **Example:** TCP for reliable connections, UDP for faster, connectionless communication like video streaming.

04 **Application Layer** :
- Includes high-level protocols used by applications for network services.
- **Example:** HTTP (for web), FTP (for file transfers), and SMTP (for email).

WHY THESE MODELS MATTER IN ETHICAL HACKING





- **Identifying Vulnerabilities:** Knowing how data flows across layers helps ethical hackers identify where security weaknesses may exist. For example, attacks like packet sniffing happen at the Data Link layer, while web-based attacks target the Application layer.
- **Troubleshooting Skills:** Understanding these models helps with troubleshooting network issues, as you can pinpoint which layer a problem might be occurring in.
- **Layer-Specific Attacks:**
  - **Physical Layer:** Physical tampering with cables or devices.
  - **Application Layer:** Exploiting software vulnerabilities through attacks like SQL injection.

TCP (TRANSMISSION CONTROL PROTOCOL)



01 **Connection-Oriented:** TCP establishes a reliable, two-way connection between client and server before data transmission begins.

02 **Reliability:** It ensures that data is received in order and without errors, using mechanisms like acknowledgments, retransmission of lost packets, and flow control.

03 **Usage:** Ideal for applications requiring data integrity, like web browsing (HTTP), file transfer (FTP), and email (SMTP).

03 **Common TCP Ports:**

- HTTP: 80
- HTTPS: 443
- FTP: 21
- SSH: 22
- SMTP: 25

UDP (USER DATAGRAM PROTOCOL)



01 **Connectionless:** UDP doesn't establish a dedicated connection, so it's faster but less reliable than TCP.

02 **Speed and Low Overhead:** UDP has no error-checking, ordering, or acknowledgment features, making it suitable for real-time applications.

03 **Usage:** Typically used in applications where speed is more important than reliability, such as video streaming, online gaming, and VoIP.

03 **Common UDP Ports:**

- DNS: 53
- DHCP: 67, 68
- SNMP: 161
- TFTP: 69
- NTP: 123

Note: DNS commonly uses UDP port 53 for general queries, but it may use TCP port 53 for larger data exchanges, such as zone transfers.

PORT AVAILABILITY IN TCP AND UDP

01 Each of the **65,536 ports** is available for both TCP and UDP, meaning there are effectively **65,536 TCP ports** and **65,536 UDP ports**.

02 Applications can use the same port number over TCP and UDP but for different purposes.



NETWORK PROTOCOLS AND PORTS



What are Network Protocols?

A network protocol is a set of rules that dictate how data is transmitted over a network. Different protocols serve different functions, such as enabling web browsing, secure connections, file transfers, and more.

Example: When you type a website address into a browser, a protocol (HTTP or HTTPS) ensures that your computer communicates with the web server hosting that site.

Common Protocols

1. HTTP (Hypertext Transfer Protocol)
2. HTTPS (Hypertext Transfer Protocol Secure)
3. FTP (File Transfer Protocol)
4. DNS (Domain Name System)
5. SSH (Secure Shell)
6. SMTP (Simple Mail Transfer Protocol)
7. POP3 (Post Office Protocol 3) and IMAP (Internet Message Access Protocol)
8. DHCP (Dynamic Host Configuration Protocol)

Importance of Knowing Ports

1. **Why Ports Matter:** Ports act as “doors” for protocols, allowing specific types of network traffic to enter or exit a device. Knowing which port a protocol uses helps in network troubleshooting, security monitoring, and managing network traffic.
2. **Vulnerable Ports:** Certain ports (like HTTP on port 80) are frequently targeted in attacks because they allow common traffic types. Monitoring traffic on these ports is crucial for network security.

PROTOCOLS AND PORT NUMBERS

Protocol	Port no.	TCP/UDP	Description	Vulnerability
FTP	21	TCP	File Transfer Protocol for transferring files (unencrypted).	Vulnerable
SSH	22	TCP	Secure Shell for encrypted remote login and data exchange.	Not Vulnerable
TELNET	23	TCP	Insecure remote terminal access (no encryption).	Vulnerable
SMTP	25	TCP	Simple Mail Transfer Protocol for sending emails.	Vulnerable
DNS	53	TCP/UDP	Domain Name System for domain-to-IP translation.	Vulnerable
DHCP	67, 68	UDP	Dynamic Host Configuration Protocol for IP addressing.	Vulnerable
TFTP	69	UDP	Trivial File Transfer Protocol (no encryption/authentication).	Vulnerable
HTTP	80	TCP	Hypertext Transfer Protocol for unencrypted web traffic.	Vulnerable
POP3	110	TCP	Post Office Protocol 3 for email retrieval (unencrypted).	Vulnerable
NTP	123	UDP	Network Time Protocol for time sync; vulnerable to DDoS amplification.	Vulnerable
IMAP	143	TCP	Internet Message Access Protocol for email (unencrypted).	Vulnerable
SNMP	161, 162	UDP	Simple Network Management Protocol for network management.	Vulnerable
LDAP	389	TCP/UDP	Lightweight Directory Access Protocol (typically unencrypted)	Vulnerable
HTTPS	443	TCP	Secure HTTP with SSL/TLS encryption for secure web browsing.	Not Vulnerable
SMB	445	TCP	Server Message Block for network file sharing (can be exploited).	Vulnerable
SMTP (SSL)	465	TCP	Secure SMTP for encrypted email transmission.	Not Vulnerable
RTSP	554	TCP/UDP	Real-Time Streaming Protocol for media streaming control.	Vulnerable
LDAPS	636	TCP/UDP	Secure LDAP with SSL/TLS for encrypted directory access.	Not Vulnerable
IMAP (SSL)	993	TCP	IMAP over SSL/TLS for secure email retrieval.	Not Vulnerable
POP3 (SSL)	995	TCP	POP3 over SSL/TLS for secure email retrieval.	Not Vulnerable
RDP	3389	TCP	Remote Desktop Protocol (can be targeted by brute-force attacks).	Vulnerable
SIP	5060, 5061	TCP/UDP	Session Initiation Protocol for VoIP; vulnerable to interception.	Vulnerable

Vulnerable means these protocols either lack encryption, have known security issues, or are commonly targeted for exploitation.

Not Vulnerable are considered secure in their encrypted versions but may still require proper configuration and management to prevent misuse.

IP ADDRESS / INTERNET PROTOCOL



An IP address is a unique identifier assigned to each device on a network. It allows devices to communicate with each other by ensuring data is sent to the correct destination.

Types of IP addresses:

01 **IPv4:** A 32-bit address format written as four numbers separated by dots (e.g., 192.168.1.1). Supports around 4.3 billion unique addresses.

02 **IPv6:** A newer 128-bit address format designed to handle the increasing number of devices, written in hexadecimal (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334). Supports a vast number of unique addresses.

IP ADDRESS / INTERNET PROTOCOL



Structure of an IPv4 Address:

01

IPv4 addresses are typically split into two main parts:

- **Network Part:** Identifies the specific network the device belongs to.
- **Host Part:** Identifies the individual device on that network.

02

Example: In the IP address 192.168.1.10, 192.168.1 could be the network part, and 10 could be the host part.

IP ADDRESS / INTERNET PROTOCOL



Note: **Class D** for Multiplexing (224-239) and **Class E** for R&D Department of Defence (240-255).

IP addresses are divided into classes to manage large, medium, and small networks. Each class has a specific range of addresses, suited for different network sizes. Now, let's see IPv4 Classes:

Class A:

- **Range:** 1.0.0.0 to 126.0.0.0
- **Uses:** Large networks, such as major corporations or ISPs (Internet Service Providers).
- **Network and Host Parts:** Uses the first 8 bits for the network part, leaving 24 bits for the host part.

Class B:

- **Range:** 128.0.0.0 to 191.255.0.0
- **Uses:** Medium-sized networks, like universities or large organizations.
- **Network and Host Parts:** Uses the first 16 bits for the network part and 16 bits for the host part.

Class C:

- **Range:** 192.0.0.0 to 223.255.255.0
- **Uses:** Small networks, often used by small businesses or homes.
- **Network and Host Parts:** Uses the first 24 bits for the network part and 8 bits for the host part.

IP ADDRESS / INTERNET PROTOCOL



Subnetting

01 Subnetting divides a network into smaller parts, called subnets, to improve efficiency and security by limiting the size of broadcast domains.

02 It Helps in efficient IP allocation and also improves Security and Performance.

03 Suppose a company with a Class B address wants to separate its departments (e.g., Sales, HR, IT) into distinct networks.

Static V/S Dynamic IP addresses

01 **Static IP Address:** An IP address manually assigned to a device, which remains the same over time and you can simply think of a Server which often use static IPs so they can be reliably accessed by other devices.

02 **Dynamic IP Address:** An IP address assigned by a DHCP server, which can change over time Like a Home routers typically assign dynamic IPs to devices connecting to Wi-Fi.

PACKET FLOW // SOURCE TO DESTINATION



Note: Every step in the packet's journey is carefully managed to ensure data integrity, reliability, and efficiency.

What is a Data Packet?

A data packet is a small unit of data that is transmitted over a network. Data is broken down into packets for efficient transfer and then reassembled at the destination.

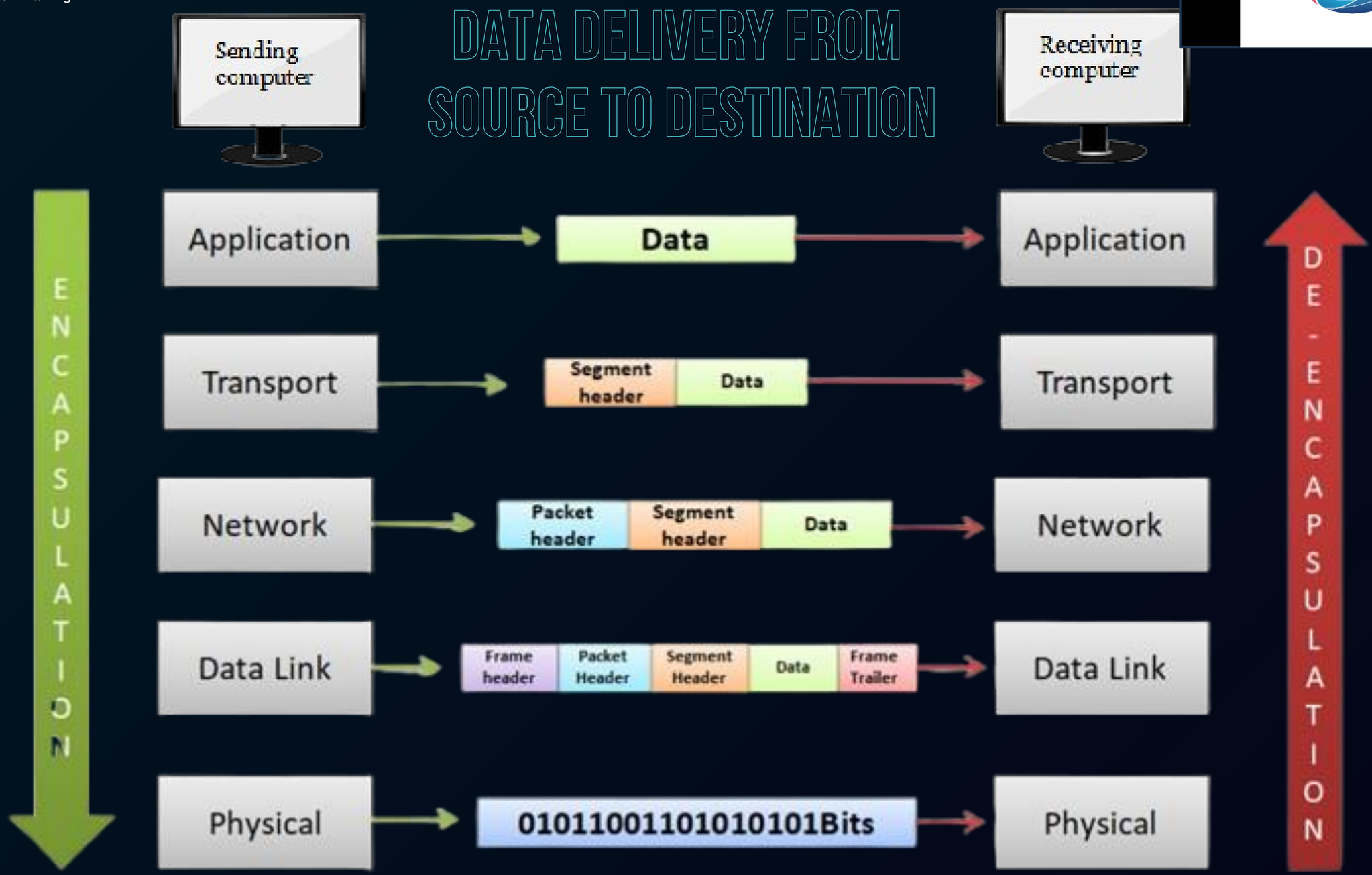
Key Steps in Packet Flow

01 **Packet Creation at Source:** The data (e.g., an email) is broken down into packets and has a header (contains information like source and destination addresses) and a payload (the actual data).

02 **Data Encapsulation:** Packets go through each OSI or TCP/IP layer, adding layer-specific information to the header it is called encapsulation like network layer = + dest. IP address, transport layer = + port number.

03 **Transmission Over Network:** Packets travel through routers, switches, and other devices on the way to their destination. Routers use the IP address in the packet header to determine the best path for the packet.

04 **Packet Reassembly at Destination:** Once packets arrive, they are reassembled in the correct order using sequence numbers in the headers. The encapsulation information is stripped away at each layer until only the original data remains, known as decapsulation.


DATA DELIVERY FROM SOURCE TO DESTINATION
Sending computer
Receiving computer
ENCAPSULATION
DE-ENCAPSULATION
Application
Data
Application
Transport
Segment header
Data
Transport
Network
Packet header
Segment header
Data
Network
Data Link
Frame header
Packet Header
Segment Header
Data
Frame Trailer
Data Link
Physical
0101100110101010101Bits
Physical

CHALLENGES & SECURITY IMPLICATION IN PACKET FLOW



01

Challenges: There are lot of challenges in packet flow which includes:

- **Packet Loss** during transmission.
- **Latency** in which it takes time for a packet to travel from source to destination.
- **Order and Reliability** in which packets may arrive out of order and protocols like TCP ensure they are reassembled correctly.

02

Security Implications in Packet Flow:

- **Packet Sniffing:** Attackers can capture data packets to steal information and tools like Wireshark are used to monitor network traffic for security analysis.

- **MITM:** Attackers intercept and alter packets during transmission, which can lead to data theft or modification. We can use and secure protocols like HTTPS and SSH helps in encrypting packets .

VULNERABLE PORTS, NETWORKS, AND VERSIONS



Vulnerable ports are commonly targeted by attackers to gain unauthorized access. Each open port can provide a way into a system if it's not properly secured.

Examples of Commonly Exploited Ports:

01 **Port 21 (FTP):** Known for insecure file transfers, as FTP transmits data in plain text.

02 **Port 23 (Telnet):** Allows remote access without encryption, making it susceptible to attacks.

03 **Port 80 (HTTP):** Used for unencrypted web traffic, which can be intercepted.

04 **Port 445 (SMB):** Often targeted in ransomware attacks; used for file sharing within networks.

VULNERABLE PORTS, NETWORKS, AND VERSIONS



Why Certain Networks Are Vulnerable

01 **Unsecured/Open Wi-Fi Networks** lack encryption, making it easy for attackers to intercept data like a public Wi-Fi network in a coffee shop can expose users to “man-in-the-middle” attacks.

02 **Misconfigured Networks** such as weak passwords or default configurations, increase vulnerability like leaving default admin credentials on a network device can allow unauthorized access.

03 **Poorly Segmented Networks** which means when all devices are on the same network segment, a compromised device can affect the entire network like a company not segmenting its internal network could allow an infected computer to spread malware to other devices.

Note: Network security depends heavily on **proper configuration, segmentation, and encryption**. Weak network configurations are prime targets for attackers.

VULNERABLE PORTS, NETWORKS, AND VERSIONS



Vulnerable Software Versions

01 **Outdated Software or Older versions of software** may have known vulnerabilities that have been fixed in newer updates like an outdated operating system or web browser may be missing critical security patches.

02 **Unpatched Vulnerabilities** of software can be vulnerable if patches for known issues aren't applied promptly like the "WannaCry" ransomware exploited unpatched vulnerabilities in older Windows operating systems.

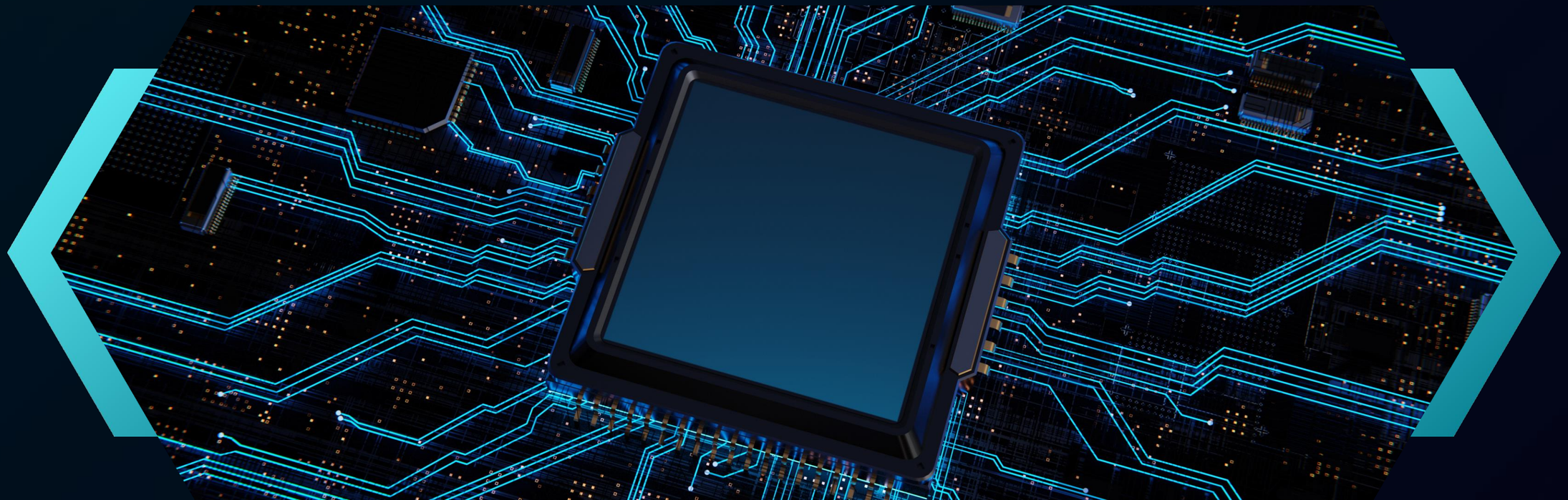
03 Keeping systems and software **up-to-date** helps protect against exploits that target known vulnerabilities like updating from older SSL/TLS versions to TLS 1.3 provides enhanced encryption, protecting data in transit.

PRACTICAL EXERCISE 1

Using Basic Networking Commands

These commands allow users to explore network configurations, monitor connections, and troubleshoot issues at various levels of detail.

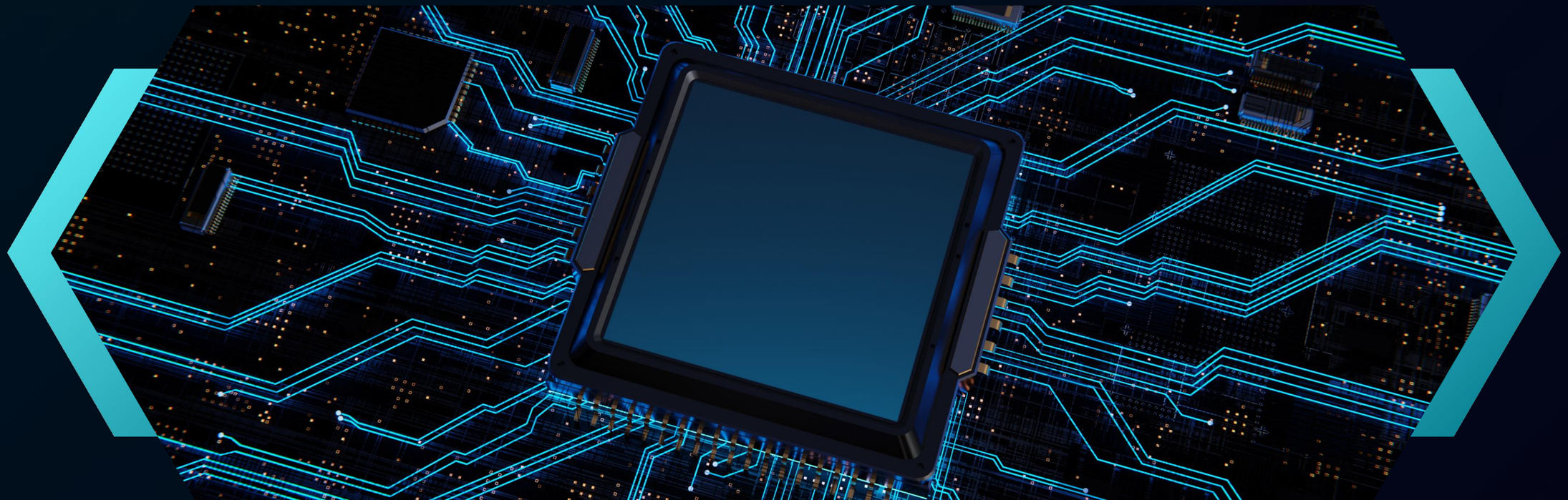
1. **ping**: Tests connectivity and measures response time.
2. **traceroute** (or **tracert** on Windows): Tracks the path to a destination.
3. **netstat**: Shows active connections and open ports.
4. **ipconfig/ifconfig**: Views and configures network interface settings.
5. **nslookup**: Resolves DNS names to IP addresses.
6. **arp**: Displays and modifies the ARP table.
7. **whois**: Retrieves registration details for a domain.



PRACTICAL EXERCISE 1 Continued

Examples of Networking Commands

1. **ping**: ping google.com
2. **traceroute/tracert**: tracert google.com
3. **netstat**: netstat -a
4. **ipconfig/ifconfig**: ipconfig or ifconfig
5. **nslookup**: nslookup google.com
6. **arp**: arp -a
7. **whois**: whois google.com



PRACTICAL EXERCISE 2

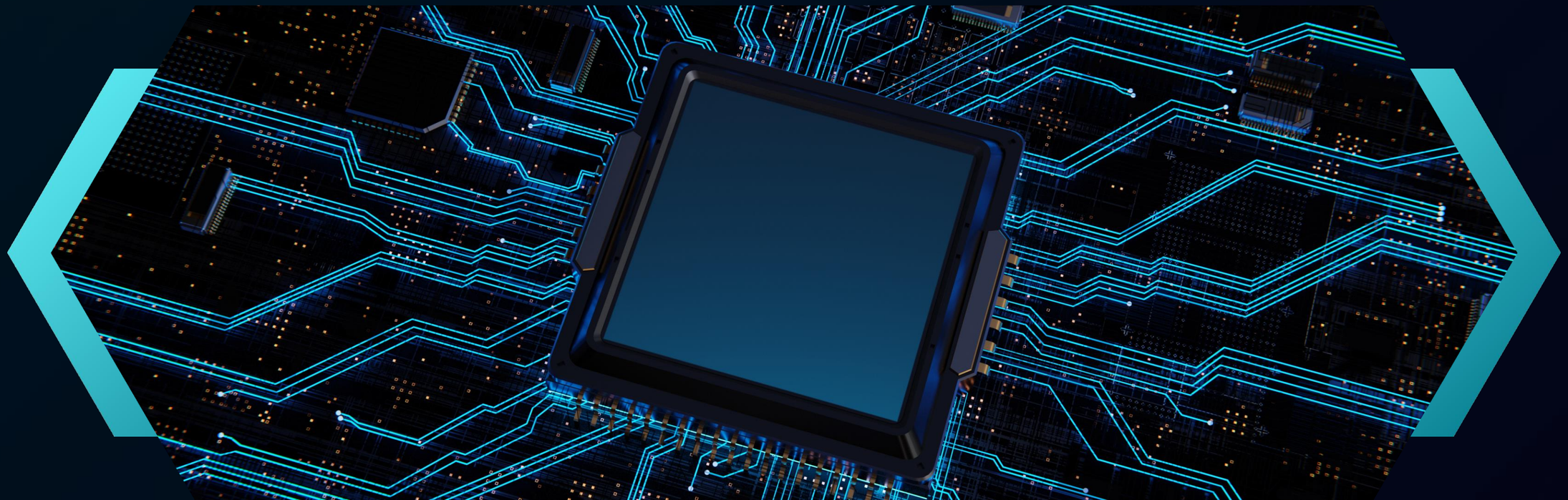
Packet Capture and Analysis with Wireshark

What is Packet Capturing?

Packet capturing is the process of **intercepting data packets** as they travel across a network. This is crucial for **monitoring network traffic** and **detecting any unusual or unauthorized data transmission.**

What is Wireshark?

Wireshark is a widely used **network analysis tool** that allows users to **capture** and **analyze packets** in real time. It's used by network professionals and ethical hackers to inspect network traffic, troubleshoot connectivity issues, and detect security threats.



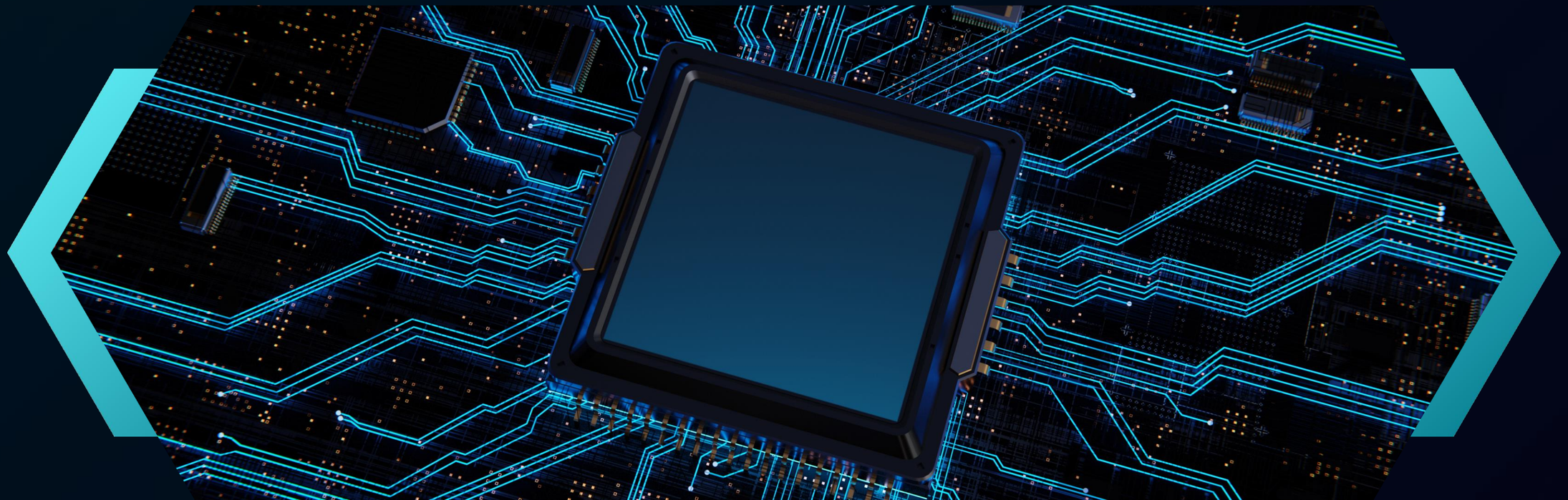
PRACTICAL EXERCISE 2 Continued

Packet Capture and Analysis with Wireshark

Installation of Wireshark: go to wireshark.org

Interface Overview

1. **Capture Interfaces:** Select which network interface to monitor (e.g., Ethernet, Wi-Fi).
2. **Packet List Pane:** Displays captured packets in a list format.
3. **Packet Details Pane:** Shows detailed information about the selected packet.
4. **Packet Bytes Pane:** Displays the raw data of the packet.





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THANK YOU

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